

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Puller jaw, Part Number 3164947
- Crankshaft gear remover, Part Number 3165049, or equivalent.

Additional Service Items

- Hone stone
- Lubriplate™ 105, or equivalent.

Preparatory Steps

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

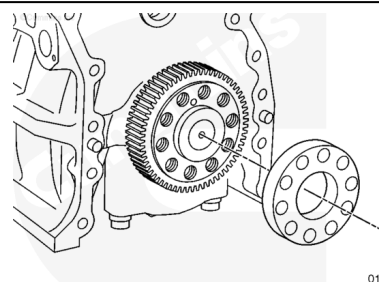
- Remove the transmission, clutch, and all related components. Refer to the equipment manufacturer's instructions.
- Remove the starter motor. Refer to Procedure 013-020 in Section 13 (</qs3/pubsys2/xml/en/procedures/89/89-013-020.html>).

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>).
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Remove the flywheel. Refer to Procedure 016-005 in Section 16 (</qs3/pubsys2/xml/en/procedures/89/89-016-005.html>).
- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16 (</qs3/pubsys2/xml/en/procedures/89/89-016-006.html>).
- Remove the camshaft idler gear. Refer to Procedure 001-036 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-036.html>).

Remove

Remove the spacer from the rear of the crankshaft.

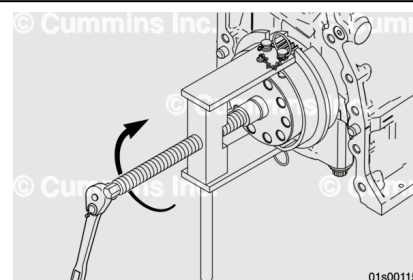
Note : If spacer can not be removed easily, proceed to removing the gear. The spacer should come off when pulling the gear.



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⚠ CAUTION ⚠

Heat can be used to aid the removal of the gear. Do not use a cutting torch. The high temperature of a torch will damage the teeth of the gear.



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⚠ CAUTION ⚠

Do not exceed 475 N•m [350 ft-lb] of torque when turning the jackscrew.

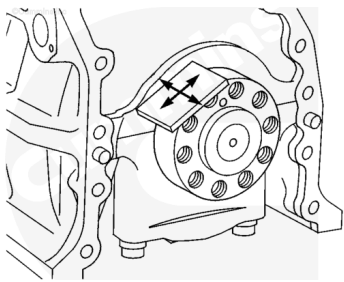
Only remove the gear when the crankshaft or gear is damaged.

Use puller jaw, Part Number 3164947. and Crankshaft Gear Remover Kit, Part Number 3165049, to remove the gear.

Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer’s recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



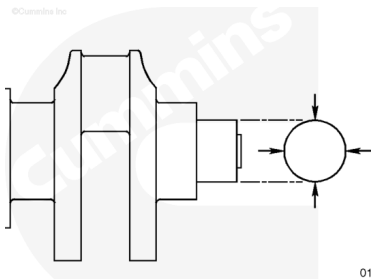
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a hone stone and polish the crankshaft outside diameter. Remove all of the small scratches, burrs, and small grooves. Use care **not** to damage the crankshaft keyway.

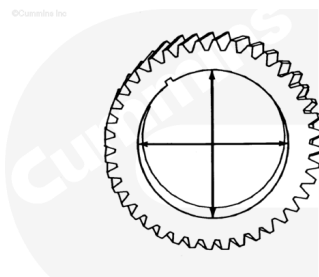
Clean and inspect the crankshaft for reuse. Measure the outside diameter.

Crankshaft Outside Diameter (Gear Location)		
mm		in
170.48	MIN	6.712
170.52	MAX	6.713



Measure the inside diameter of the rear crankshaft gear.

Rear Crankshaft Gear Inside Diameter		
mm		in
170.335	MIN	6.706
170.375	MAX	6.708



If the crankshaft gear is **not** within specifications, the crankshaft gear **must** be replaced.

Install

⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.

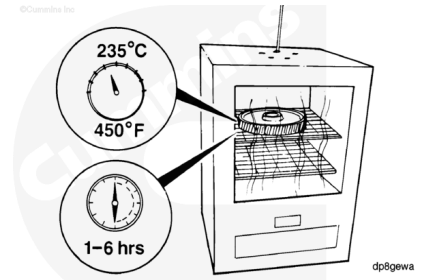
⚠ CAUTION ⚠

Do not exceed the specified time or the temperature. Damage to the gear and gear teeth will result.

⚠ CAUTION ⚠

Do not attempt to install the gear without using heat.

Use an oven and adjust the heat to 235°C [450°F]. Heat the gear for a minimum of 1 hour, but **not** more than a maximum of 6 hours. The inside diameter of the gear will become larger and simplify installation.

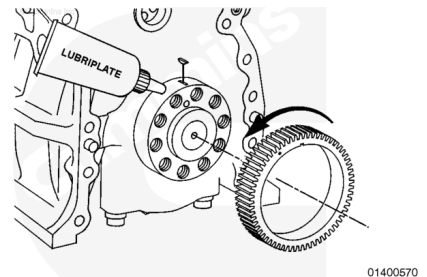


⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.

⚠ CAUTION ⚠

Allow the gear to cool slowly. Do not use water or oil to reduce the cooling time. This will cause the gear to crack.



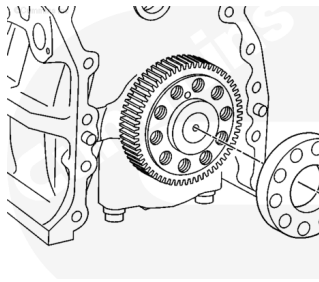
The timing mark on the crankshaft gear **must** be visible from the front of the gear after it is installed on the crankshaft.

Install a new key into the crankshaft.

Lubricate the outside diameter of the crankshaft with Lubriplate™ Number 105, or equivalent. Remove the gear from the oven and install it on the crankshaft.

Install the crankshaft spacer onto the rear of the crankshaft.

Use two flywheel bolts and washers to pull the spacer onto the crankshaft nose.



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Finishing Steps

⚠ WARNING ⚠

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- Install the flywheel housing. Refer to Procedure 016-006 in Section 16 (</qs3/pubsys2/xml/en/procedures/89/89-016-006.html>).
- Install the flywheel. Refer to Procedure 016-005 in Section 16 (</qs3/pubsys2/xml/en/procedures/89/89-016-005.html>).
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Fill the lubricating oil system. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>) in Section 7 for engine oil capacity.
- Install the starter motor. Refer to Procedure 013-020 in Section 13 (</qs3/pubsys2/xml/en/procedures/89/89-013-020.html>).
- Install the camshaft idler gear. Refer to Procedure 001-036 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-036.html>).
- Install the transmission, clutch, and all related components. See equipment manufacturer service information.